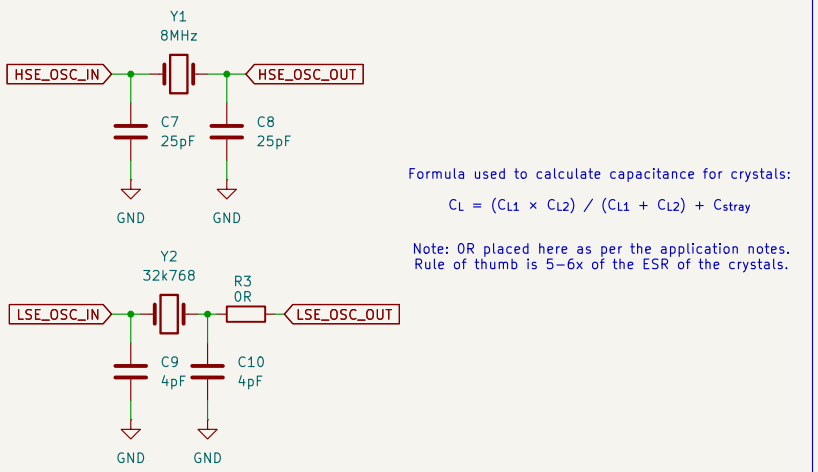
STM32F103C8T6 Overview

RCC Crystals

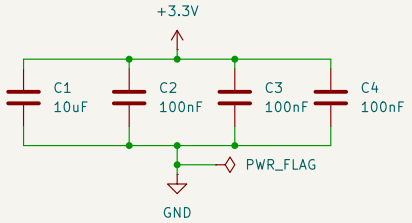


Formula used to calculate capacitance for crystals:

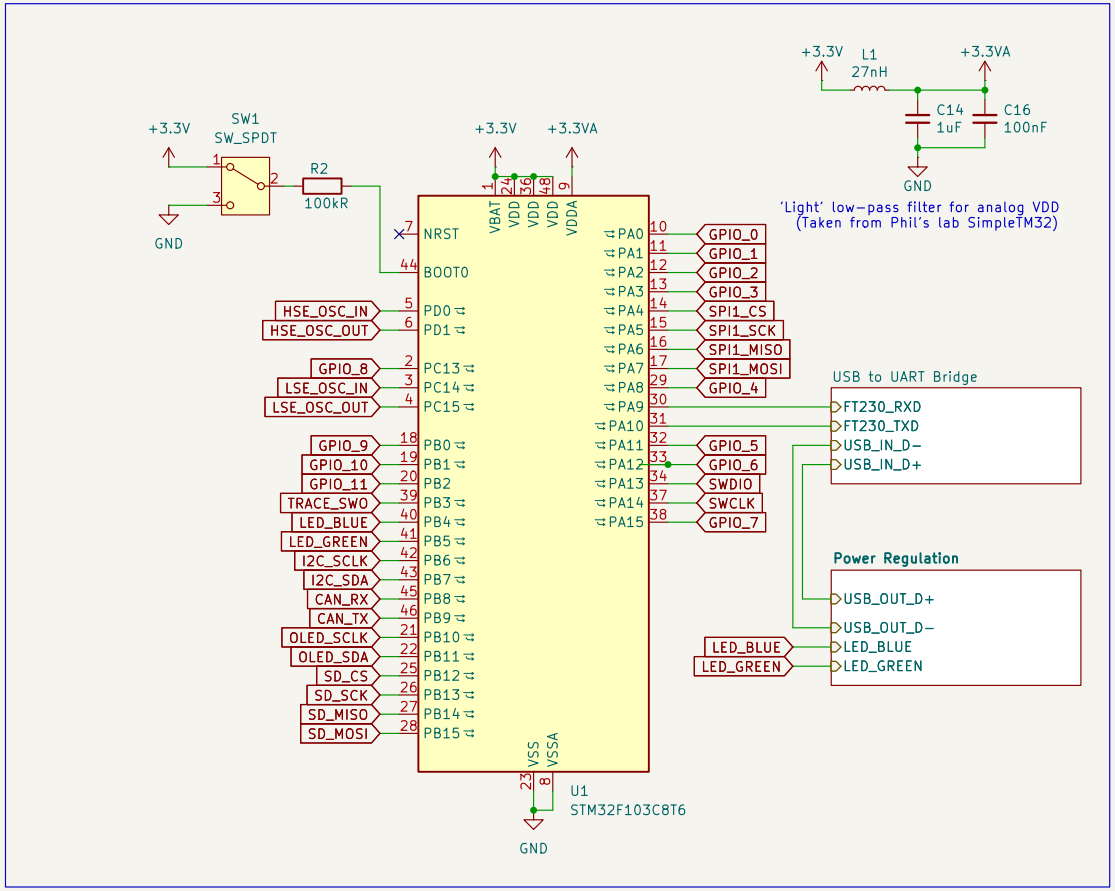
$C_L = (C_{L1} \times C_{L2}) / (C_{L1} + C_{L2}) + C_{stray}$

Note: 0R placed here as per the application notes.
Rule of thumb is 5–6x of the ESR of the crystals.

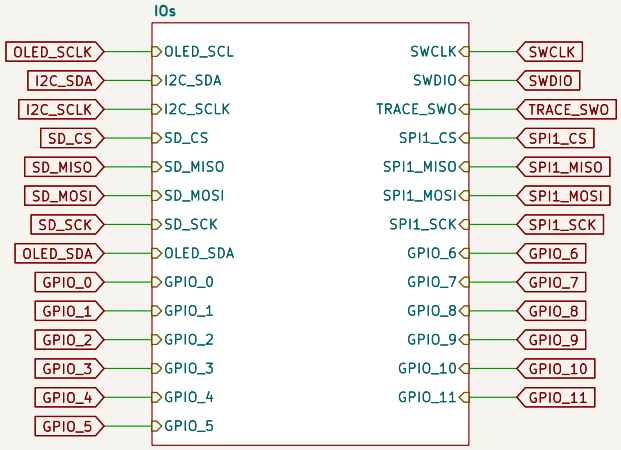
Decoupling Capacitor



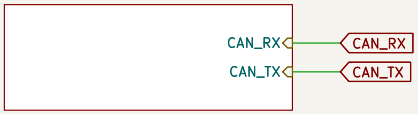
The number of 100nF capacitor reflects to the number of available VDD pins.
As per the application notes:
In addition, each power supply pair must be decoupled with filtering ceramic capacitors C (100 nF) and a chemical capacitor C of about 10 µF connected in parallel on the STM32F10xxx device.



IOs



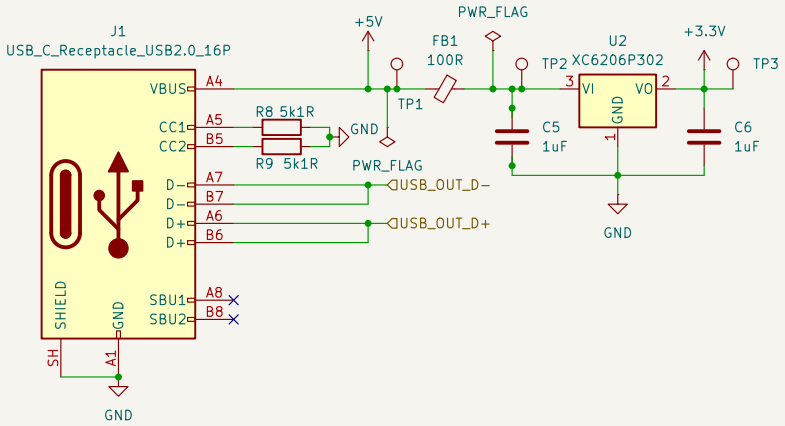
CAN Transceiver



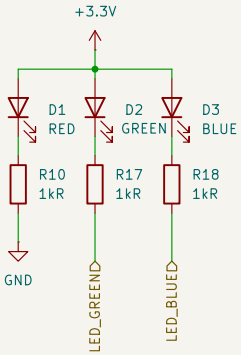
Sheet: /owl_v1/
File: owl_v1.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 1/6



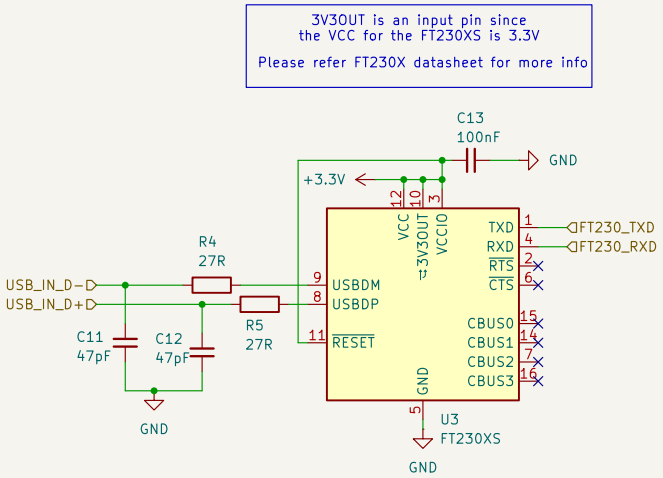
Regulates power from USB−C 5V to 3.3V via XC6206P302 power regulator.
Ferrite beads are placed to reduce noises at high frequencies.
Inspiration taken from Phil's lab STM32 design
(https://github.com/pms67/STM32F1−Breakout−Board)



Copied schematics from here:
https://github.com/fred314159265/CANchovy

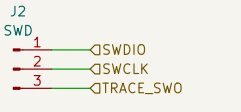
RED − indicates board is receiving 3.3V power
GREEN − User programmable
BLUE − User programmable

Sheet: /owl_v1/Power Regulation/ File: usb_power_regulator.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 2/6

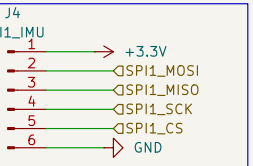


Sheet: /owl_v1/USB to UART Bridge/		
File: usb_to_uart.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 3/6

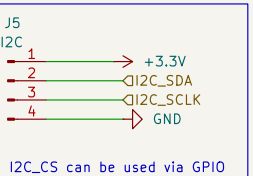
Serial Wire Debugger



SPI-IMU breakout

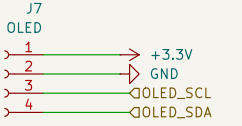


I2C Breakout



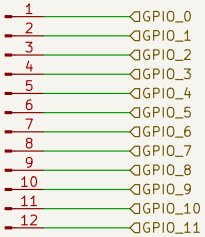
I2C_CS can be used via GPIO

OLED Display Module Integration

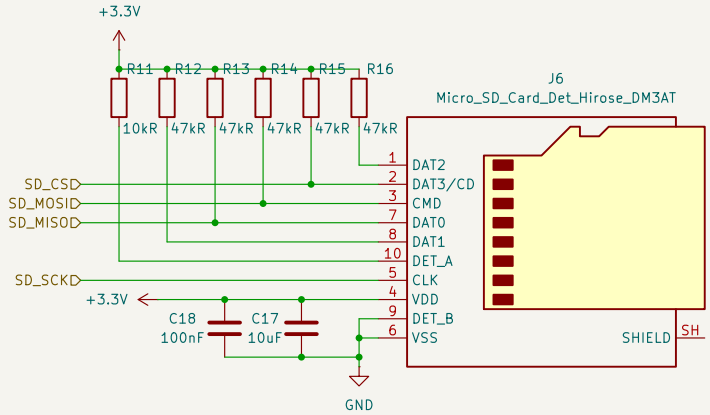


I2C Display to use
0.96" OLED Display Module (128x64)

External GPIOs



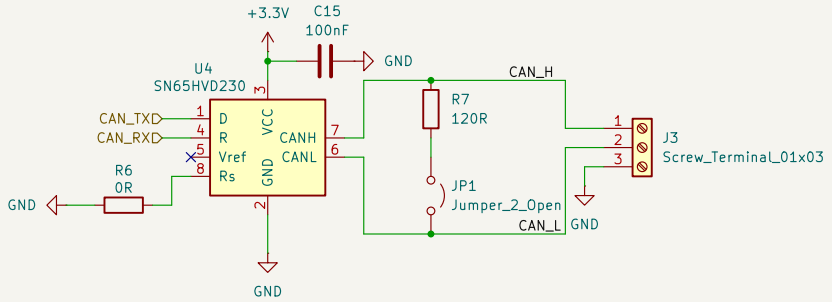
SD Card Logging via SPI



A good schematic design reference by KaiPereira:
https://github.com/KaiPereira/CM5-SODIMM-Carrier/blob/master/PCB/SCHEMATIC.pdf

Also, for the SPI to SD card pinouts, I followed this:
https://forum.armbian.com/topic/12736-orangepi-zero-external-sd-cad/

Sheet: /owl_v1/IOs/ File: GPIOs.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 4/6



Sheet: /owl_v1/CAN Transceiver/		
File: can_controller.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 5/6

TODO:
- Integrate external power input other than USB power
- LEDs where necessary?

Sheet: /Design Notes/
File: design_notes.kicad_sch

Title:

Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.7 | Id: 6/6